IIT ROPAR TIF – AWADH APPLICATION PRESENTATION

Software, Web & Application Development Portfolio

Applicant: **OM RAJ** | Category: **Software / Application Development / Website**

Live Application & Store Links:

- SNIFF DOI 2.0 Live on Microsoft Store: View Microsoft Store App Page
- IIT Ropar AWE LAB Official Portal: View Live Portal
- Personal Website: https://om1904.github.io/omraj-website/

1. SNIFF DOI 2.0 — Academic Literature Discovery Engine

Live on Microsoft Store

250M+ Federated Article Index

Built a high-throughput search engine enabling cross-disciplinary reference retrieval, open-access resolution, and citation extraction.

6-Tier Research Gap Engine

Automates theoretical deficit analysis, experimental bottlenecks, connected paper citation graphs, and Q1 target journal recommendations.

CSL Citation Engine

Implemented automated citation formatting engine supporting APA 7th, IEEE, Harvard, Vancouver, and BibTeX export formats.

Microsoft Store Distribution

Packaged and published as a native Windows desktop workstation app. [View MS Store Page]

2. WASH Suite v2.0 — Environmental Chemistry Solvers

Live on Microsoft Store

APHA & BIS Calculation Solvers

Programmed physicochemical calculation algorithms for pH, TDS, TSS, BOD5, COD, alkalinity, and heavy metal concentrations based on APHA 24th Ed & BIS 10500.

Linear Regression Calibration

Built spectrophotometer multi-point linear calibration module with standard curve fitting and absorbance validation.

NABL ISO-17025 PDF Exporter

Automated generation of official laboratory PDF compliance certificates for STP/ETP facility managers.

Live Store Distribution

Live on Microsoft Store for environmental engineers and water quality laboratories. [View MS Store Page]

3. Institutional Web Portals & Web Development

IIT Ropar & Web Apps

AWE LAB Official IIT Ropar Portal

Designed, authored, and launched the official academic research laboratory web portal for Dr. Indramani Dhada's group at IIT Ropar. [View Live Portal]

Multi-Page Portfolio Web App

Built responsive, light-theme academic website equipped with interactive lightboxes, live WQI calculator, search filters, and embedded Google Forms. [View Website]

Frontend UI/UX Architecture

Clean HTML5, CSS3, JavaScript (ES6+), custom theme variables, dynamic layout math, and mobile responsiveness.

Automated Node.js Tooling

Experienced in writing Node.js build scripts, REST API integrations, and automated headless PDF rendering workflows.

4. Value Proposition & Alignment with AWaDH TIF

IIT Ropar iHub – AWaDH

Full-Stack Software Capability

Ready to design and maintain AWaDH web portals, dashboard tools, MIS systems, and startup repository portals.

Proven Store & Web Track Record

4 published Microsoft Store applications and live academic web portals demonstrate end-to-end execution capabilities.

ICPS & DeepTech Familiarity

Background at IIT Ropar combines technical software engineering with environmental technology, Cyber-Physical Systems, and data modeling.

Immediate On-Site Availability

Already associated with IIT Ropar, ready to contribute to AWaDH software, web, and technology projects immediately.